rezki ::= engineer | educator

Rezki Boureghdad

Back-end & Data Architect · Computer Science Educator

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PROFILE

Software engineer focused on back-end and data architecture, and a computer-science educator. At Solrus, an Algiers software studio, I design the data models, APIs and integrations behind client production platforms. I also teach at Université M'Hamed Bougara, where I have covered 12 subjects, from algorithms and databases to compilation and semantics, at Bachelor's and Master's level. I take the same approach in both roles: get the structure right, make it measurable, and explain it clearly.

EXPERIENCE

Data Specialist, Structure & Architecture

2025 – Present

Solrus — software studio for AI, cloud and secure systems · Algiers, remote-first

- Own back-end and data architecture for client platforms: domain modelling, relational schemas and migrations, API contracts, and third-party integrations.
- Build event-driven and async Python services (FastAPI, SQLAlchemy 2, Kafka, Redis, Celery) and tune them for latency and reliability.
- Security in the design from the start: role-based access, 2FA, audit trails, data-residency compliance, hardening before audits.
- Test-first delivery with pytest against real PostgreSQL and end-to-end Playwright suites. Work closely with front-end, AI and product teammates.

> Most client projects are under NDA. Names and details are not disclosed.

Freelance Full-Stack Developer (auto-entrepreneur)

May 2021 – Present

Independent · web, mobile and AI products

- Run projects end to end for local businesses and community organisations: scoping, proposal, clickable prototype, build and deployment.
- Member of the team behind **Zakat.business**, an award-winning non-profit startup project.

Teaching Assistant — Computer Science

Nov 2024 – Present

Université M'Hamed Bougara, Boumerdès

- 12 subjects taught so far to Software Engineering and LMD students, Bachelor's and Master's level.
- Prepare and run tutorials and labs, write exercises and assessments, and help students connect theory to hands-on practice.
- Adjust my teaching for each audience, from first programming courses to formal semantics for Master's students.

Development Mentor

Oct 2024 – Sep 2026

DevUp Club, Université M'Hamed Bougara

- Designed and ran practical software-engineering workshops for students, with hands-on material that participants applied to real projects.

EDUCATION

MSc Software Engineering & Data Science
UMBB · Jan 2024

BSc Computer Science
UMBB · Jan 2022

SUBJECTS TAUGHT

Algorithms & Data Structures

Databases

Programming Languages (intro → advanced)

Compilation

Semantics

+ 7 other CS modules

CERTIFICATIONS

NVIDIA
Fundamentals of Deep Learning · Intro to Machine Learning

Neo4j
Graph Data Science · Fundamentals · Graph Data Modeling

EF SET English
73/100 — C2 Proficient

LANGUAGES

Arabic
Native

English
C2 (EF SET)

French
Professional working

SELECTED PROJECTS

Public, open and personal work only.

Forsa / Aamal ATS — recruitment platform

Solrus public case study

- Connects companies, independent recruiters and job seekers in Algeria. Semantic job matching with sentence-transformer embeddings and location search with PostGIS.
- WebSocket chat, community forum, FCM push notifications, SlickPay subscriptions, JWT/Argon2 auth. Flutter mobile app and admin panel.

FastAPI · PostgreSQL/PostGIS · Flutter · Docker

e-motion — back-office for cash-on-delivery shops

Own product

- Orders, stock, couriers and expenses, with real profit per order. Weighted-average purchase cost, stock that can never go below zero, alerts for customers who often cancel or return, and an audit log.
- Excel import, packing slips, English/French interface. Runs on free hosting tiers.

Next.js · TypeScript · Supabase · Vercel

VoidSketches — free CS lessons through analogies

Own product

- Animated, narrated lessons in English and French, one concept each, starting with compilation. Local TTS pipeline whose timings drive animation and captions.

React · Vite · TTS · voidsketches.vercel.app

DocMCP — local PDF search server for AI tools

Open source

- Ingests PDF folders into ChromaDB and serves them to any MCP client (Kiro, Claude Code) as semantic-search tools. Fully local: ONNX / sentence-transformer embeddings, LLM on Ollama.
- Recursive Language Model reasoning: a local LLM explores the full documents in a Python REPL instead of relying on top-k chunks. Web UI and one-command Docker setup.

Python · Node.js · MCP · ChromaDB · Ollama · Docker · github.com/V0idSeeker/RAG-MCP

Community apps — Ahl Quran & 7alkaty

Community

- **Ahl Quran:** multi-tenant Quran-school manager for study circles, attendance and progress (FastAPI, PostgreSQL, Flutter).
- **7alkaty:** mosque course and imam-following app with scheduled push notifications (Node/Express, MongoDB, Expo).

Forest-fire detection & reporting

Master's thesis

- Co-designed a three-part system: a citizen reporting app, a data platform and alerting. Detection models built with YOLO and Faster R-CNN.

Python · YOLO · R-CNN · Flutter · Firebase

claim-bot — migration-number Discord bot

Own tool

- Reserves the next migration number before the file exists, so developers never collide. Merges committed files from the GitHub API with open claims, and checks the Alembic chain for extra heads, broken links and cycles.

Python · discord.py · GitHub GraphQL · Alembic

ISO 27001 skill for Claude Code

Tooling

- Checks a codebase against ISO/IEC 27001:2022, plans the rollout step by step, and checks every edit with a scanner hook.

Python · Security compliance · AI tooling

SKILLS

Back-end
Python, FastAPI, Django/DRF, SQLAlchemy 2 (async), Alembic, Pydantic, Celery, Node/Express

Data
PostgreSQL/PostGIS, Redis, Kafka, MongoDB, Neo4j, Elasticsearch, Supabase

AI / ML
RAG, MCP servers, LangGraph, FAISS/Chroma, embeddings & reranking, LLM APIs, YOLO/R-CNN

Front & mobile
TypeScript, Next.js, React, Tailwind, Flutter/Dart, React Native

Ops & quality
Docker, nginx, GitHub Actions, Vercel, Grafana/Loki, pytest, Playwright

Security
JWT, Argon2, OAuth2, TOTP 2FA, ISO 27001, data residency

Also
Java, C++, SQL; contract-first API design, decision records, AI-assisted development

CS FOUNDATIONS

Algorithms & complexity

Compilers & formal languages

Formal methods

Deep learning

Programming-language semantics

Database theory & graph databases